The Paleomagnetism of the New Jersey Triassic: A Field Study of the Inclination Error in Red Sediments

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Abstract. It has previously been reported that on redeposition in the laboratory the magnetization of both red sediments and varves has a systematic error in inclination which could be a serious source of error in paleomagnetic work. To test whether such an error occurs in nature, the directions of magnetization measured in specimens from intrusions, extrusions, and red sediments of the Newark system have been compared. There is no significant difference between the stable directions in each type of rock, showing that no important error in inclination occurs in the sediments.

INTRODUCTION

The Newark system in the New Jersey–Pennsylvania area contains extensive exposures of the three types of rock that have been widely used for paleomagnetic work—red beds (from which preliminary results have already been reported by *Du Bois, Irving, Opdyke, Runcorn, and Banks* [1957]), basic intrusives, and lava flows, all of which are closely associated in time and space. It is thus possible to compare the directions of magnetization in the three types of rock and, further, to compare the over-all results with those previously obtained from Triassic beds of North America.

One of the main uncertainties in paleomagnetic work in sediments is the question of the inclination error. In the laboratory it has been shown that in the redeposition of varves [*King*, 1955; *Griffiths, King, Rees, and Wright*, 1960] and red sediments [*Clegg, Almond, and Stubbs*, 1954] the inclination is several degrees lower than that of the applied field, whereas the agreement in declination is very good. Such inclination errors could be due to the preferential settling of elongated particles or postdepositional compaction.

The main question is, do these two effects (deposition and compaction errors) occur in nature? The best test is to compare direction observed in igneous rocks with those of sediments of the same age, because such error will not be present in the former. If the inclination is systematically lower in the sediments than in igneous rocks, it would seem that such errors are present. Before this test can be carried out it is necessary to provide evidence that the directions observed are those which were acquired at the time of formation and that the directions have not been changed since. Strong evidence of stability is therefore needed.

OUTLINE OF THE GEOLOGY OF THE BEDS STUDIED

The Triassic of the east coast of North America occurs in a series of structural basins from Nova Scotia to North Carolina. The New Jersey Triassic forms part of the biggest of these basins, which extends from the Hudson River to northern Virginia (Fig. 1). The sediments and igneous rocks of these basins have been subjected to block tilting and gentle warping. The average dips are 10° to 12°, usually to the northwest, and they rarely exceed 20°.

The sediments of the Newark group in New Jersey are continental in origin and reach a thickness of more than 17,000 feet along the Delaware River. They were initially divided into three formations [*Kummell*, 1897] called, in ascending order, the Stockton, the Lockatong, and the Brunswick. More recent work [*McLaughlin*, 1944, 1946; *McLaughlin and Willard*, 1949] has shown that these formations are different facies which are synchronous in part. The Stockton lithofacies is typically light-colored coarse-grained arkosic sandstone and

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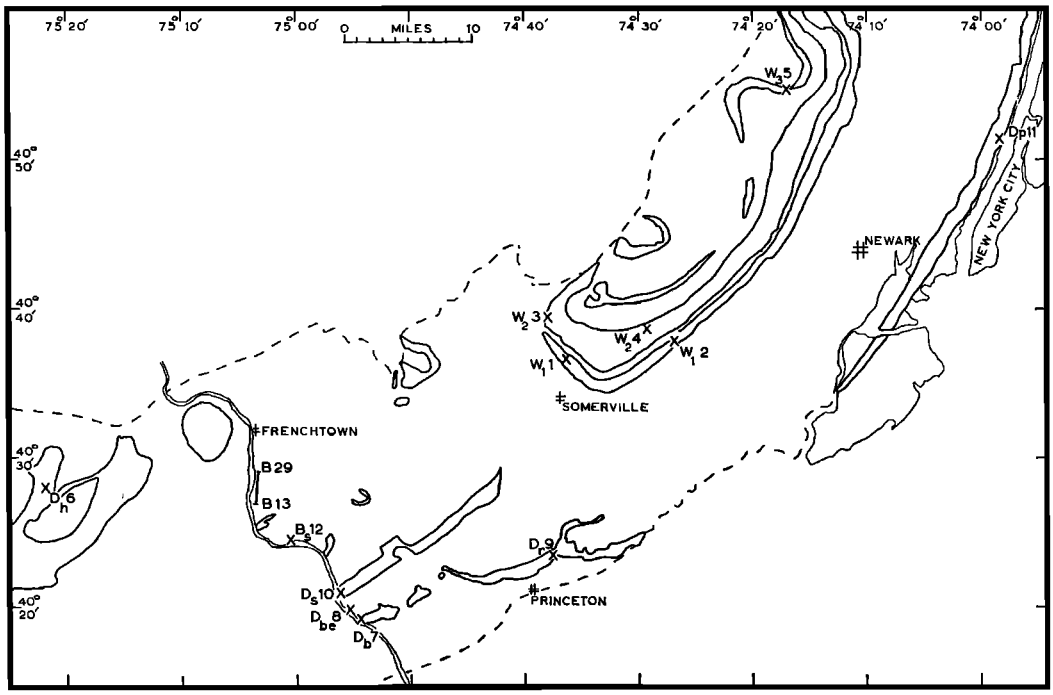


Fig. 1. Locality sketch-map. The collecting localities are marked with crosses and labeled as in Table 1. The Triassic sedimentary outcrop is dotted and the igneous outcrop in full line. The intrusive and extrusive outcrops may be distinguished since the localities in the former are labeled D and in the latter W.

conglomerate, red to brown fine-grained sandstone, and red siltstones. The Lockatong lithofacies consists mainly of dark gray to black argillite. Red shales, sandstones, and argillites constitute the Brunswick lithofacies. The age of these sediments is uncertain, owing to the inconclusive nature of the paleontological evidence. It would seem that the most probable age is that put forward by McLaughlin [*Reeside and others*, 1957, p. 1493], who states that 'for the present it seems best to adopt a somewhat arbitrary view that the Newark is roughly the equivalent of the greater part of the Keuper, or Upper Triassic.'

Associated with these sediments are dolerite intrusives and basalt flows (Fig. 1) which are chemically identical and differ from one another in texture only. The basalts, known as the Watchung flows, are contained in sediments of the Brunswick lithofacies near the top of the Newark group. The flows are multiple in character. The dolerite intrusives occur as dikes and sills which, it is thought, were intruded into the sediments near the end of the Newark deposition. The intrusives cannot be older than the sediments which they intrude nor younger than the Cretaceous, since they were faulted, warped, and eroded before the Cretaceous overlap. Recently, potassium-argon dates obtained from the Palisades sills and Watchung flows by *Erickson and Kulp* [1960] placed their age at 190 million years, that is, Triassic.

Field Survey

Samples were oriented with a Brunton compass to an accuracy of about 3°. Cores, 3.5 cm in diameter, were drilled from the samples, and several disk specimens were cut from these. The directions of magnetization were then measured to an accuracy of 3° with an astatic magnetometer, the methods employed being similar to those described by *Collinson, Creer, Irving, and Runcorn* [1957].

Sediments. A section of the Newark group sediments was sampled along the Delaware River valley between Stockton and Frenchtown

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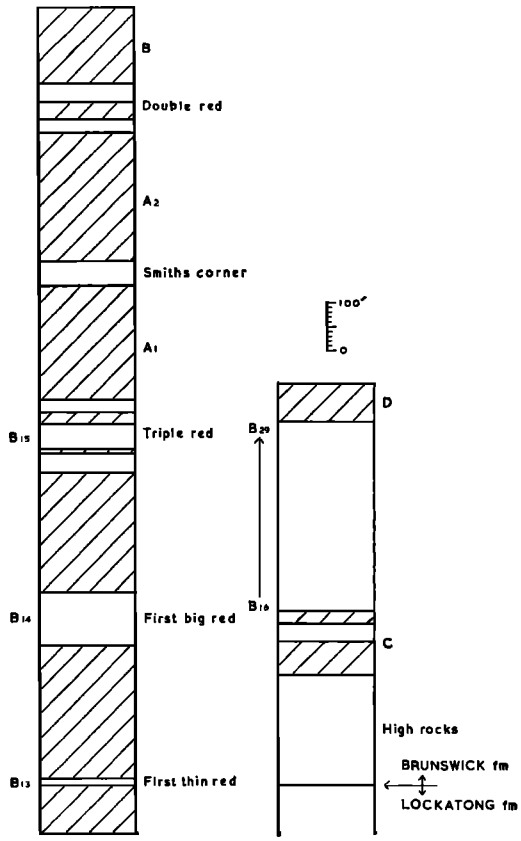


Fig. 2. Stratigraphic spacing of samples in the Brunswick lithofacies. The terminology is McLaughlin's. There is no stratigraphic break in this section.

from rock of the Brunswick lithofacies (B13-29). These specimens were collected from fresh road cuttings on New Jersey Route 29a. One sample (B_s12) from the Stockton formation was taken from a quarry above Stockton, on Route 29a. Figure 2 shows the stratigraphic position of the samples. Five samples were collected from the 'First Thin Red' (B13), four from the 'First Big Red' (B14), and two from the middle member of the 'Triple Red' (B15). Thirteen samples (sites B16–B28) were collected at intervals of approximately 20 feet from the base to within 100 feet of the top of the rock unit between McLaughlin's units C and D. Results from the upper 100 feet of this unit were reported by *Du Bois, Irving, Opdyke, Runcorn, and Banks* [1957] and will be used in this paper as a single site (B29). Figure 3 shows the results for all disks measured.

Intrusions. Two or more oriented samples were collected from each site (usually quarries or road cuttings) in the lavas and intrusives. The intrusive samples were taken from the Palisades sill at Rocky Hill (D_r9) and at the George Washington Bridge (D_p11), the Sourland Mountain sill below Lambertville, New Jersey (D_s10), the small intrusions at Belle Mountain ($D_{bs}8$) and Baldpate Mountain (D_b7) above Titusville, New Jersey, and the Haycock Sill north of Quakertown, Pennsylvania (D_h6) (Fig. 1). The initial measurements gave a mean direction $D = 350.5°$, $I = +34.6°$.

Lava flows. Samples from the Watchung flows were taken at five localities. There are two in the first flow (W_1)—one north of Somerville in an abandoned quarry half a mile southeast of the old copper mine (W_11) and the other on the Stoney Brook Road half a mile north of North Plainfield (W_12). There are two in the second flow (W_2)—one south of Far Hills on the road from Route 202, just south of north branch of the Raritan, to Liberty Corner (W_23) and the other east of Mt. Bethel in a quarry along Cory's Brook (W_24). The last site is in the third flow (W_3) at Mountain View west of the Pompton River (W_35). The initial

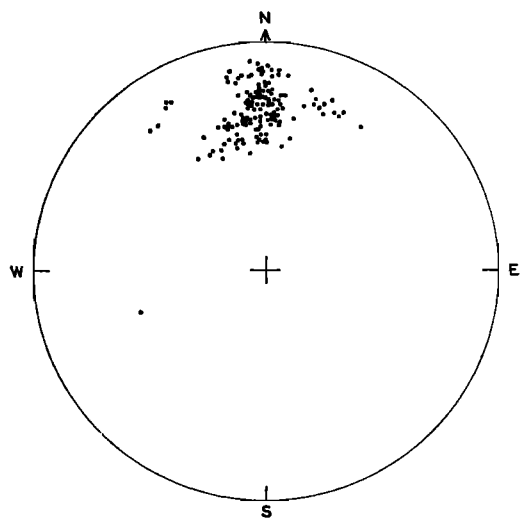


Fig. 3. Directions of natural remanent magnetization observed in sedimentary specimens. The directions have been corrected for geological dip so that the plane of the projection (an equal-area projection) is the bedding plane at each site. The inclination scale is given in Figure 6.

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TABLE 1. Coercivity of Remanence and the Isothermal Saturation Remanent Magnetization

Unit	H_s	H_{cr}	M_s
Lavas			
3rd flow	1500	200	275
2nd flow	1250	380	350
1st flow	1250	200	370
Intrusives			
Baldpate Mt.	1250	250	260
Rocky Hill	1200	280	550
Belle Mt.	1000	275	395
Sourland sill	1500	240	192
Haycock sill	1500	260	400

Notes: H_s is the steady field required to induce the saturation isothermal remanent magnetization (M_s). H_{cr} is the back field required to destroy M_s. Fields are measured in oersteds and moments in emu/cm^3 $\times$ 10^{-3}.

measurements gave a mean direction $D = 7.3$, $I = +37°$.

Magnetic properties (Table 1). Specimens from the three lava flows and the intrusive sites were subjected to steady magnetic fields in order to determine the coercivity of remanence (H_{cr}) and the isothermal saturation remanent magnetization (M_s). The fields required to saturate the specimens ranged between 1000 and 1500 oersteds. The coercivity of remanence ranged between 200 and 380 oersteds. Several sedimentary specimens were also treated in the same manner. In a peak field of 11,600 oersteds the remanence had not yet become saturated. The back field needed to nullify the moment induced in this field was 5400 oersteds. Such a high value is presumably due to the presence of finely divided hematite, and testifies to the high magnetic stability of these rocks.

Stability

The directions of natural remanent magnetization from many of the igneous sites showed 'streaking'; that is, the directions in different specimens are along a great circle between the present earth's field and the direction of original magnetization. This is due to the presence of a component induced by the present earth's field, and, in order to minimize the effect of this, component specimens were treated in alternating magnetic fields of 150-oersted peak strength [*Brynjolfsson*, 1957; *As and Zijderveld*, 1958; *Creer* 1958, 1959] in accordance with the criteria established by *Irving, Stott, and Ward* [1960]. This led to a marked improvement in the concentration of the vectors for all localities that had shown 'streaking' (Fig. 4). For the localities that had not shown streaking the concentration in most cases was increased, but in some it was decreased slightly. In these cases the mean declination and inclination did not change significantly. Table 2 gives the mean directions and dispersions for samples from the various sites before and after treatment. The directions for all specimens of the lavas and intrusives after treatment in 150 oersteds are plotted in Figures 5 and 6. Figure 7 gives a typical demagnetization curve for a lava specimen containing a large secondary component imposed by the present field.

Sedimentary specimens were also subjected to alternating field demagnetization; however, there was no significant change in direction in fields up to 900 oersteds (peak) and only a 5 per cent decrease in intensity (Fig. 7). *Creer* [1957, 1959] found a greater decrease in intensity at these fields in unstable Keuper marls.

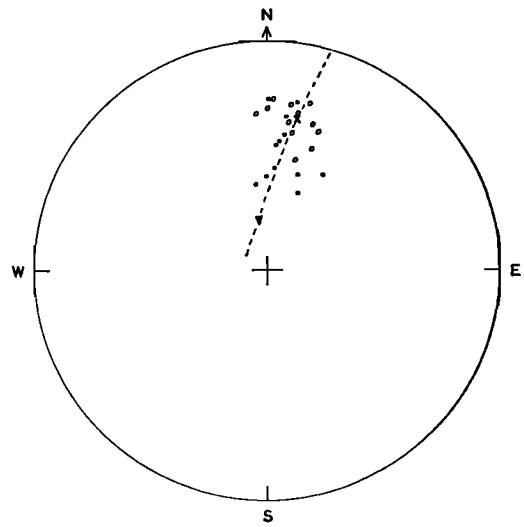


Fig. 4. Directions of magnetization at site W_s5 before (dots) and after (circles) treatment in an alternating magnetic field of 150 oersteds (peak). The inclination is much decreased by this treatment (see Table 2). The dotted great circle is drawn through the present field (black triangle) and the mean stable direction (a cross) of the specimens after treatment. These points are not corrected for geological dip.

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TABLE 2. Mean Directions and Dispersions for Samples from the Various Sites

LOCALITY			Strike,	Dip,	M_n						$M_{(150)}$			
Site	Lat.	Long.	degrees	degrees	S	N	D	I	R	k	D	I	R	k
Lavas														
W_1 1	40°36′	74°36′	322	12NE	3	12	12	35	11.67	33	3	34	11.87	85
W_1 2	40°38′	74°27′	48	12NW	2	8	355	24	7.96	175	3	23	8.84	50
W_2 3	40°40′	74°38′	18	12E	3	10	17	35	9.88	75	18	25	11.93	157
W_2 4	40°30′	74°30′	55	10NW	2	8	7	23	7.93	100	19	14	7.98	350
W_3 5	40°55′	74°15′	14	12N	3	12	4	57	11.56	25	10	31	11.79	52
Intrusives														
D_h 6	40°28′	75°28′	20	15NW	3	13	351	26	12.76	50	360	11	12.44	21
D_b 7	40°18′	74°53′	30	15NW	3	12	358	47	11.88	92	4	36	11.87	85
D_{be} 8	40°20′	74°55′	35	16NW	3	12	333	33	11.68	34	339	31	11.65	31
D_r 9	40°22′	74°38′	66	18N	3	12	352	24	11.38	18	354	25	11.59	27
D_s 10	40°21′	74°56′	40	18NW	2	8	7	34	7.91	78	358	31	7.92	88
D_p 11	40°51′	73°58′	24	12W	3	12	358	42	11.68	34	4	30	11.85	73
Sediments														
B_s 12	40°25′	75°W	68	13N	3	21	359	33	20.75	80				
B 13	40°28′	75°4′W	72	14N	5	33	351	38	32.67	97				
B 14	40°28′	75°4′W	65	9NW	4	19	4	29	18.66	53				
B 15	40°28′	75°4′W	59	13NW	2	12	347	36	11.12	12				
B 16	40°28′	75°4′W	65	12N	1	9	359	25	8.96	200				
B 17	40°28′	75°4′W	65	12N	1	5	359	25	4.98	200				
B 18	40°28′	75°4′W	65	12N	1	3	357	15	...	...				
B 19	40°28′	75°4′W	65	12N	1	6	4	17	5.97	167				
B 20	40°28′	75°4′W	65	12N	1	5	358	21	4.98	200				
B 21	40°28′	75°4′W	65	12N	1	5	327	19	4.98	200				
B 22	40°28′	75°4′W	65	12N	1	1	359	25	...	...				
B 23	40°28′	75°4′W	65	12N	1	2	356	17	...	...				
B 24	40°28′	75°4′W	65	12N	1	2	356	15	...	...				
B 25	40°28′	75°4′W	65	12N	1	6	24	25	5.98	250				
B 26	40°28′	75°4′W	65	12N	1	2	351	23	...	...				
B 27	40°28′	75°4′W	65	12N	1	6	360	24	5.48	10				
B 28	40°28′	75°4′W	65	12N	1	3	352	21	...	...				
B 29	40°28′	75°4′W	65	12N	21	71	6	28	...	...				

Notes: Site details and attitude of beds or attitudo of adjacent sediments at the igneous sites are given in the first five columns. S is the number of samples and N the number of specimens. The values in the columns headed $\mathbf{M}_n$ and $\mathbf{M}_{(150)}$ refer to the natural remanent magnetization and the magnetization after treatment in 150-oersted alternating field. D and I are the declination and inclination of the mean site direction corrected for geologic dip; R is the resultant and k is the measure of within-site precision [*Fisher*, 1953]. In the column headed $\mathbf{M}_{(150)}$, N is 9 for site 2 and 12 for site 3, because three original data sheets were misplaced and it is impossible to repeat measurements after treatment.

Clearly the natural remanent magnetization of these sediments is very stable.

Unfortunately, it was not possible to demonstrate stability decisively by using the fold test [*Graham*, 1949] because of the low structural dips. When correction for geological dip is made, however, the scatter does become less; for instance, the improvement in the Watchung flows after correction to the dip of adjacent sediments is from a k of 73 to one of 94, and in the intrusives the k increases from 27 to 38. These changes are not significant at $P = 0.05$, but are in the right sense for stability. The k used is *Fisher's* [1953] dispersion parameter estimated from the mean site directions obtained after treatment in alternating fields of 150 oersteds.

The directions of natural remanent magnetization in the sediments (Fig. 3) and the directions in the igneous rocks after treatment (Figs. 5 and 6) show a close grouping around a mean direction which differs from the present

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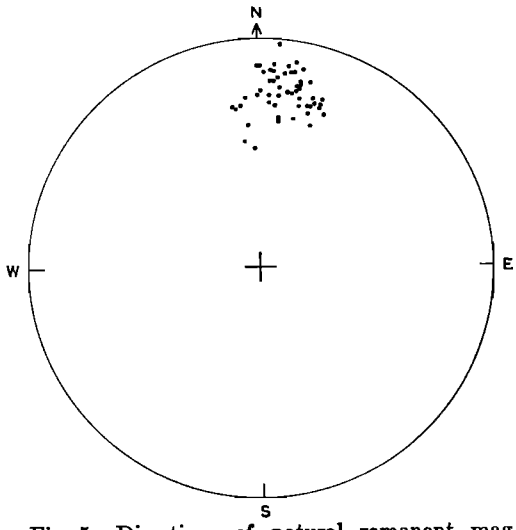


Fig. 5. Directions of natural remanent magnetization observed in the Watchung Lavas. The directions have been corrected to the bedding of adjacent sediments (values given in Table 2). The inclination scale is given in Figure 6.

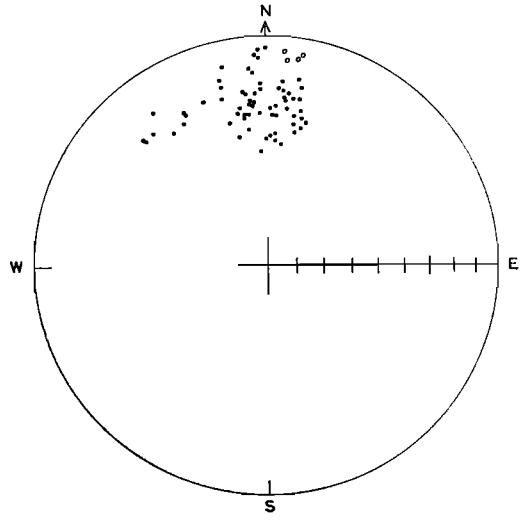


Fig. 6. Directions of natural remanent magnetization observed in the dolerite intrusives. The directions have been corrected to the bedding of adjacent sediments (values given in Table 2.) The inclination scale is given. The solid points indicate north seeking pole downwards, open circles north seeking pole upwards.

earth's field by about 50°. This uniformity of directions through a very considerable stratigraphic thickness in different types of rock is strong evidence of the directions observed in Figures 3, 5, and 6 being of very high stability.

Analysis

The directions used in the analysis are the directions of natural remanent magnetization in the sediments and the directions in the igneous rocks after treatment in a 150-oersted alternating field. The mean for each type of rock computed as a means of sites is given in Table 3. Tests [*Watson*, 1956] were applied to the mean direction for each site, and it was found that there was no significant difference ($P = 0.05$) between the dispersions or between the mean directions for the three rocks. The over-all mean for the Triassic of New Jersey is therefore calculated by giving each site unit weight (Table 3).

Discussion

Inclination error. In artificially deposited varves *King* [1955] found that, if the applied field was inclined at an angle of 25°, the error in inclination was about 10°. Comparison of Figures 3, 5, and 6, the data given in Tables 2 and 3, and the analysis referred to above shows that there is no such difference between the sediments and igneous rocks studied in this paper. It is therefore concluded that in the Newark group sediments an inclination error of paleomagnetic significance is not present.

Magnetostriction. If pressure effects had any permanent influence, they must have affected both igneous rocks and sediments similarly. In view of their varied histories this seems un-

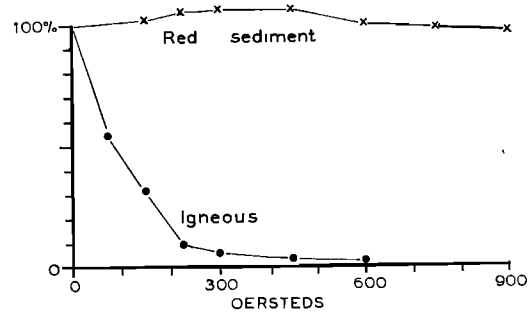


Fig. 7. Demagnetization curves. The maximum field applied is just under 900 oersteds. The ratio of the moment after treatment to the original moment is plotted vertically. The upper curve (crosses are experimental points) is from a red sediment showing the extreme stability. The lower curve is for a partially unstable lava.

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TABLE 3. Mean Directions for Different Types of Rock

Rock	*B*	*D*	*I*	*R*	*k*
Newark sediments	18	357.2	+23.5	17.63	46
Dolerite intrusions	6	355.8	+27.7	5.87	38
Watchung flows	5	8.6	+23.5	4.96	100
New Jersey Triassic	29	358.7	+25.1	28.36	44

Notes: *B* is the number of sites, *D* is the declination and *I* the inclination of the mean direction, *R* is the resultant and *k* the precision [*Fisher*, 1953]. The 95 per cent error for the New Jersey Triassic is 4.1°.

likely. In fact, it is difficult to see how this agreement could have originated in any other way than by alignment along the earth's field at the time of deposition.

Reversals. In the rocks sampled there are no reversals of direction of magnetization. *Du Bois* [1957] reported a reversal from the Connecticut Triassic, and reversed zones are also known from the Triassic of the western United States [*Collinson and Runcorn*, 1960]. The stratigraphic position of the reversal in Connecticut is not known to the writer, but it is obvious that if a reversed zone could be found in the New Jersey Triassic the problem of correlating the New Jersey and Connecticut sections with each other and with those of the western United States would be greatly simplified.

Paleomagnetic pole positions. Table 4 gives the pole positions derived from the New Jersey Triassic and from various other Triassic formations in North America. The variation in longitude of the poles from the North American Triassic is 24° and in latitude, 15°. The poles all fall in what is now Siberia, in the neighborhood of Lake Baikal (Fig. 8), in very good agreement with the poles obtained from formations separated by distances of the order of 2500 miles.

Pole positions have been computed from other formations by several workers. *Graham* [1955] published results from several Triassic formations, the Redonda, the Chinle, and the New Oxford. Graham did not calculate poles from these formations, and it is doubtful that he ever meant this to be done since in many cases insufficient thicknesses of sediments were sampled to insure against errors due to secular variation. Poles were calculated by others from his figures, however, notably by *Cox and Doell* [1960], and information concerning these sites appears in their review. *Kintzinger* [1957] also did not calculate poles; however, one of his sites is given above (Moenkopi from Marble Canyon) after *Cox and Doell* [1960]. His other site in the Dinosaur Canyon Sandstone member of the Moenave formation was rejected on the basis that an insufficient thickness of sediment was sampled. The other important result is from Nova Scotia lavas by Bowker [*Cox and Doell*, 1960]. Since these results have appeared in this review only on the basis of a personal letter, it would seem that the best course would be to wait until the work appears in full before using it. *Runcorn* [1956] and *Collinson and*

TABLE 4. Triassic Paleomagnetic Poles from the United States

Formation	Sampling Area		Pole Position				Reference
	Lat.	Long.	Lat.	Long.	$d\chi$	$d\psi$	
Moenkopi	Utah and Arizona		62N	103E	12½	12½	*Cox and Doell* [1960]
Chugwater	Wyoming		48N	112½E	5	5	*Collinson and Runcorn* [1960]
Connecticut Triassic	42N	73W	55N	88E	15	8	*Du Bois and others* [1957]
Connecticut valley lavas, Massachussets	42N	73W	55N	88E	11	6	*Irving and Banks* [1961]
Newark group (N. J.)	40.5N	74.9W	63N	108E	4	3	This paper

Notes: $d\chi$ and $d\psi$ are the errors at right angles to and in the direction of colatitude. The pole for the Moenkopi formation was calculated by *Cox and Doell* [1960] from data published in *Collinson and Runcorn* [1960], *Kintzinger* [1957], and *Runcorn* [1956].

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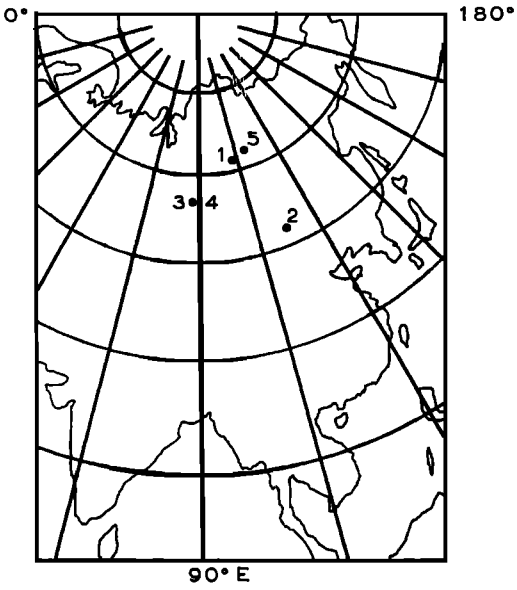


Fig. 8. Paleomagnetic poles for the Triassic of the U.S.A. The poles are numbered as in Table 4. Poles 3 and 4 coincide. The map is a stereographic projection of part of the eastern half of the northern hemisphere. These poles are of course *relative* to the U.S.A. *only;* the continental outlines are added here for comparative purposes only.

Runcorn [1960] took many samples from the Chinle formation; however, most of the sites showed streaking effects due to magnetic instability and were not used. The one site that has been published from the Springdale formation [*Runcorn,* 1956] also shows streaking due to instability, and was also rejected as a basis of comparison. The results given in Table 4 are considered by the author to be the most reliable.

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References

As, J. A., and J. D. A. Zijderveld, Magnetic cleaning of rocks in paleomagnetic research, *Geophys. J., 1,* 308–319, 1958.

Brynjolfsson, A., Studies of remanent magnetism and viscous magnetism in the basalts of Iceland, *Advances in Phys., 6,* 247–254, 1957.

Clegg, J. A., M. Almond, and P. H. Stubbs, The remanent magnetism of some sedimentary rocks in Britain, *Phil. Mag., 45,* 583–598, 1954.

Collinson, D. W., K. M. Creer, E. Irving, and S. K. Runcorn, Paleomagnetic investigations in Great Britain, 1, The measurement of the permanent magnetization of rocks, *Phil. Trans. Roy. Soc. London A, 250,* 73–82, 1957.

Collinson, D. W., and S. K. Runcorn, Polar wandering and continental drift: Evidence from paleomagnetic observations in the United States, *Bull. Geol. Soc. Am., 71,* 915–958, 1960.

Cox, A., and R. R. Doell, Review of paleomagnetism, *Bull. Geol. Soc. Am., 71,* 645–768, 1960.

Creer, K. M., Paleomagnetic investigations in Great Britain, 5, The remanent magnetization of unstable Keuper marls, *Phil. Trans. Roy. Soc. London A, 250,* 130–143, 1957.

Creer, K. M., Preliminary paleomagnetic measurements from South America, *Ann. géophys., 14,* 373–390, 1958.

Creer, K. M., A.C. demagnetization of unstable Triassic Keuper marls from S.W. England, *Geophys. J., 2,* 261–275, 1959.

Du Bois, P. M., Comparison of paleomagnetic results for selected rocks of Great Britain and North America, *Advances in Phys., 6,* 177–186, 1957.

Du Bois, P. M., E. Irving, N. D. Opdyke, S. K. Runcorn, and M. R. Banks, The geomagnetic field in Upper Triassic times in the United States, *Nature, 180,* 1186–1187, 1957.

Erickson, G. P., and J. L. Kulp, Potassium-argon measurements on the Palisades diabase and associated basalts (Abstract), *J. Geophys. Research, 65,* 2487–2488, 1960.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London A., 217,* 295–306, 1953.

Graham, J. W., The stability and significance of magnetism in sedimentary rocks, *J. Geophys. Research, 54,* 131–167, 1949.

Graham, J. W., Evidence of polar shift since Triassic time, *J. Geophys. Research, 60,* 329–347, 1955.

Griffiths, D. H., R. F. King, A. I. Rees, and A. E. Wright, The remanent magnetism of some recent varved sediments, *Phil. Trans. Roy. Soc. London A, 256,* 359–383, 1960.

Irving, E., and M. R. Banks, Paleomagnetic results from the Upper Triassic lavas of Massachusetts, *J. Geophys. Research, 66,* 1935–1939, 1961.

Irving, E., P. M. Stott, and M. Ward, Demagnetization of igneous rocks by alternating magnetic fields, *Phil. Mag.,* in press.

King, R. F., The remanent magnetism of artificially deposited sediments, *Monthly Notices Roy. Astron. Soc., Geophys. Suppl. 7,* 115–134, 1955.

Kintzinger, P. R., Paleomagnetic survey of Triassic rocks from Arizona, *Bull. Geol. Soc. Am., 68,* 931–932, 1957.

Kummel, H. B., *Annual Report of the State Geol-*

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ogist for the Year 1896, New Jersey Geol. Surv., 25–88, 1897.

McLaughlin, D. B., Triassic stratigraphy in the Point Pleasant district, Pennsylvania, *Pa. Acad. Sci. Proc., 18,* 62–69, 1944.

McLaughlin, D. B., The Triassic rocks of the Hunterdon Plateau, New Jersey, *Proc. Penna. Acad. Sci., 20,* 89–98, 1946.

McLaughlin, D. B., and B. Willard, Triassic facies in the Delaware Valley, *Proc. Penna. Acad. Sci., 23,* 34–44, 1949.

Reeside, J. B., and others, Correlation of the Triassic formations of North America, *Bull. Geol. Soc. Am., 68,* 1451–1514, 1957.

Runcorn, S. K., Paleomagnetic survey in Arizona and Utah, preliminary results, *Bull. Geol. Soc. Am., 67,* 301–316, 1956.

Watson, G. S., Analysis of dispersion on a sphere, *Monthly Notices Roy. Astron. Soc., Geophys. Suppl., 7,* 153–159, 1956.

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